PLaID:Preference Aligned Language Model for Targeted Inorganic Materials Design

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Motivation

- Discovering new materials is critical for technologies like solar cells, batteries, and carbon capture.
- Traditional discovery relies on slow and expensive trial-and-error experimentation.
- Chemical space is extremely large; only a fraction of possible materials have been explored.
- Generative AI models can help explore new materials more efficiently.

Goal of the Paper

- Develop a generative AI system capable of producing stable crystal structures.
- Use large language models (LLMs) to represent and generate inorganic materials.
- Align the model to prefer thermodynamically stable materials.
- Improve rates of Stable, Unique, and Novel (S.U.N.) materials.

Key Idea: PLaID

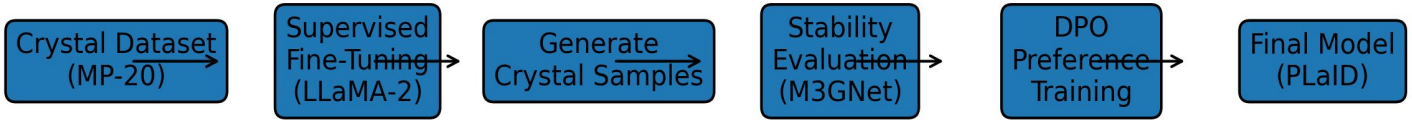
- PLaID = Preference Aligned Language Model for Inorganic Design

- Pipeline:

  1. Fine-tune LLaMA-2 on crystal structure data
  2. Represent crystals using symmetry-aware text format (Wyckoff representation)
  3. Apply Direct Preference Optimization (DPO) to bias the model toward stable crystals

PLaID Model Training Pipeline

PLaID Training Pipeline



Why Language Models?

- LLMs can treat crystal structures as text sequences.
- Flexible prompting enables multiple tasks:
  - Structure generation
  - Infilling
  - Conditional design
- Pretrained models require fewer training samples.

Crystal Symmetry and Wyckoff Positions

- Crystal structures have symmetry defined by space groups.
- Wyckoff positions describe atomic sites using symmetry constraints.
- Instead of listing every atom position, symmetry operations generate them.
- The paper encodes these symmetry concepts directly as text tokens.

Wyckoff Text Representation

- The model represents each crystal as a text sequence including:
  - Chemical formula
  - Space group
  - Lattice parameters
  - Atomic elements
  - Fractional coordinates
  - Wyckoff site symmetries

This improves structural validity and generation efficiency.

Direct Preference Optimization (DPO)

- DPO aligns model outputs with preferred outcomes.
- Training pairs: stable crystal vs unstable crystal.
- Stability determined using M3GNet predictions.
- The model learns to prefer stable structures during generation.

Dataset and Training Setup

Dataset: MP-20 dataset

- 45,231 inorganic materials
- Structures with up to 20 atoms

Model:

- LLaMA-2 7B
- LoRA fine-tuning
- 4-bit quantization
- PyTorch + Transformers framework

Evaluation Metrics

1. Validity – physical plausibility of structures
2. Coverage – diversity compared to real materials
3. Property distribution – similarity to dataset statistics
4. Stability – energy above convex hull
5. S.U.N Rate – Stable, Unique, Novel materials

Experimental Results

- PLaID significantly improves stability and S.U.N rate.
- Best configuration: Wyckoff representation + DPO.
- Achieves ~60% higher stability rate compared to prior models.
- Achieves ~40% higher S.U.N rate.

Example Performance Numbers

Best PLaID Model:

- Stability Rate (relaxed): 20.5%
- Unique: 74%
- Novel: 94%
- Final S.U.N Rate: 14.2%

Experimental Results Table

Table 1: Results for unconditional materials generation on the MP-20 dataset. Shows ablated results for PLaID, where base refers to LLaMA-2 7B using a coordinate representation and no DPO.

Method	Params	Validity (%) (↑)		Coverage (%) (↑)		Property Dist. (↓)		Stability (%) (↑)		S.U.N. (%) (↑)
		Structural	Composition	Recall	Precision	d_ρ	d_{elem}	Initial	Relaxed	
CDVAE	–	99.93	86.93	98.31	99.35	0.914	1.654	0.1	3.6	3.5
DiffCSP	–	99.61	82.23	99.53	99.35	0.257	0.403	8.9	12.5	9.7
DiffCSP++	–	**99.99**	85.81	99.48	**99.66**	0.278	0.408	8.9	13.2	9.1
FlowMM	–	96.43	83.37	99.47	99.71	0.291	**0.079**	4.1	9.3	6.3
SymmCD	–	94.32	85.85	**99.64**	98.87	**0.090**	0.399	5.0	9.4	7.0
CrystalLLM 7B†	–	96.4	**93.3**	91.1	94.9	3.61	1.06	–	–	–
LLaMA-2 7B										
PLaID-Types	Base	99.21	87.98	97.56	99.55	0.824	0.084	6.3	17.0	9.2
	DPO	98.40	88.91	96.99	99.56	1.325	0.165	7.9	**20.9**	12.2
	Wyckoff	99.87	91.28	98.32	99.64	0.129	0.110	**11.4**	17.5	12.0
PLaID	Wyckoff & DPO	99.67	93.03	98.07	99.44	1.500	0.102	8.6	20.5	**14.2**

† Our CrystalLLM reproduction, PLaID Base, significantly outperforms reported CrystalLLM results. We explore this in Appendix A.2.

Why This Matters

- Traditional random structure search finds <1% stable materials.
- PLaID dramatically improves discovery efficiency.
- Demonstrates power of adapting NLP alignment methods to scientific discovery.

Future Work

- Conditional generation for targeted properties
- Scaling to larger models and datasets
- Validate generated structures using Density Functional Theory (DFT)
- Real-world materials discovery applications

Conclusion

- Introduces symmetry-aware text representation for crystals.
- Applies Direct Preference Optimization to materials design.
- Achieves state-of-the-art results in generating stable, novel materials.
- Shows LLM alignment techniques can accelerate scientific discovery.

Thank You!